Knowing Unknowns in an Age of Incomplete Information

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Background

Digitizing Human Lives

- Scale
  - $8.5B$ web searches per day
  - $40M$ books digitized $\rightarrow 130M$
- COVID-19 –
  - 47% rise in broadband usage
  - ~~32%~~ 62% parents report teens' daily internet use exceeding four hours

Digitizing Human Lives

Concern 1

- Misinformation: Information consumed $\neq$ ground truth
- Bias: [Information consumed $\neq$ ground truth] + discriminates against a social group

- Ground truth?

[1] Flanagin & Metzger, 2000
[2] Tucker & Persily, 2020
[3] Bail, 2021

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- Misinformation: Information consumed $\neq$ ground truth
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- Ground truth?
  - ~~A norm~~ An exception on the Internet[1]
  - Consider two statements:
    - S_1: The earth is flat ✗
    - S_2: People think the earth is flat ✓

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- Ground truth?
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  - Consider two statements:
    - S_1: The earth is flat ✗
    - S_2: People think the earth is flat ✓
    - S_1 and S_2 have similar effect on the reader[2]
    - Telling the reader S_1 is False doesn't help either[3]

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[3] Bail, 2021

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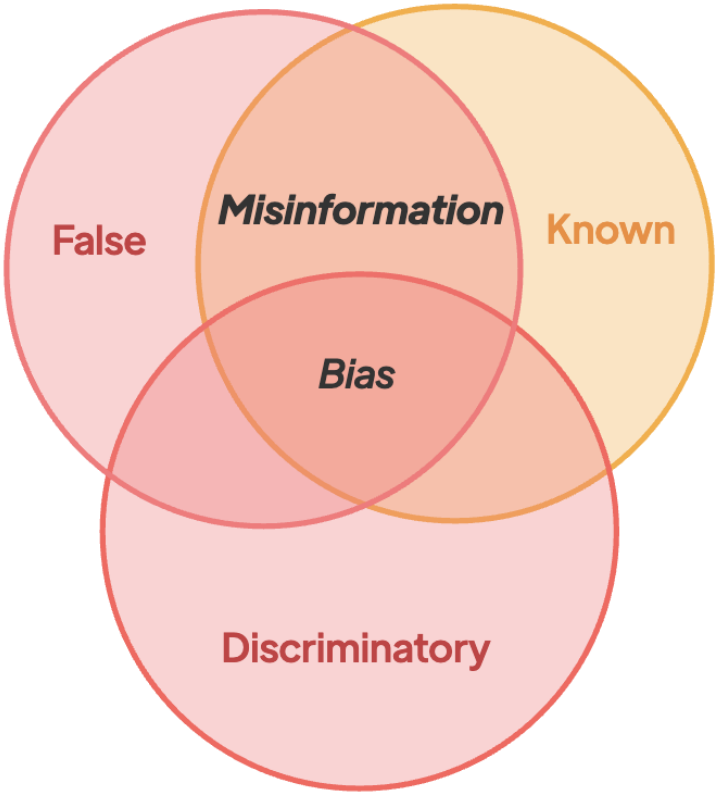
Concern 2

Information Space on the Internet



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Concern 2

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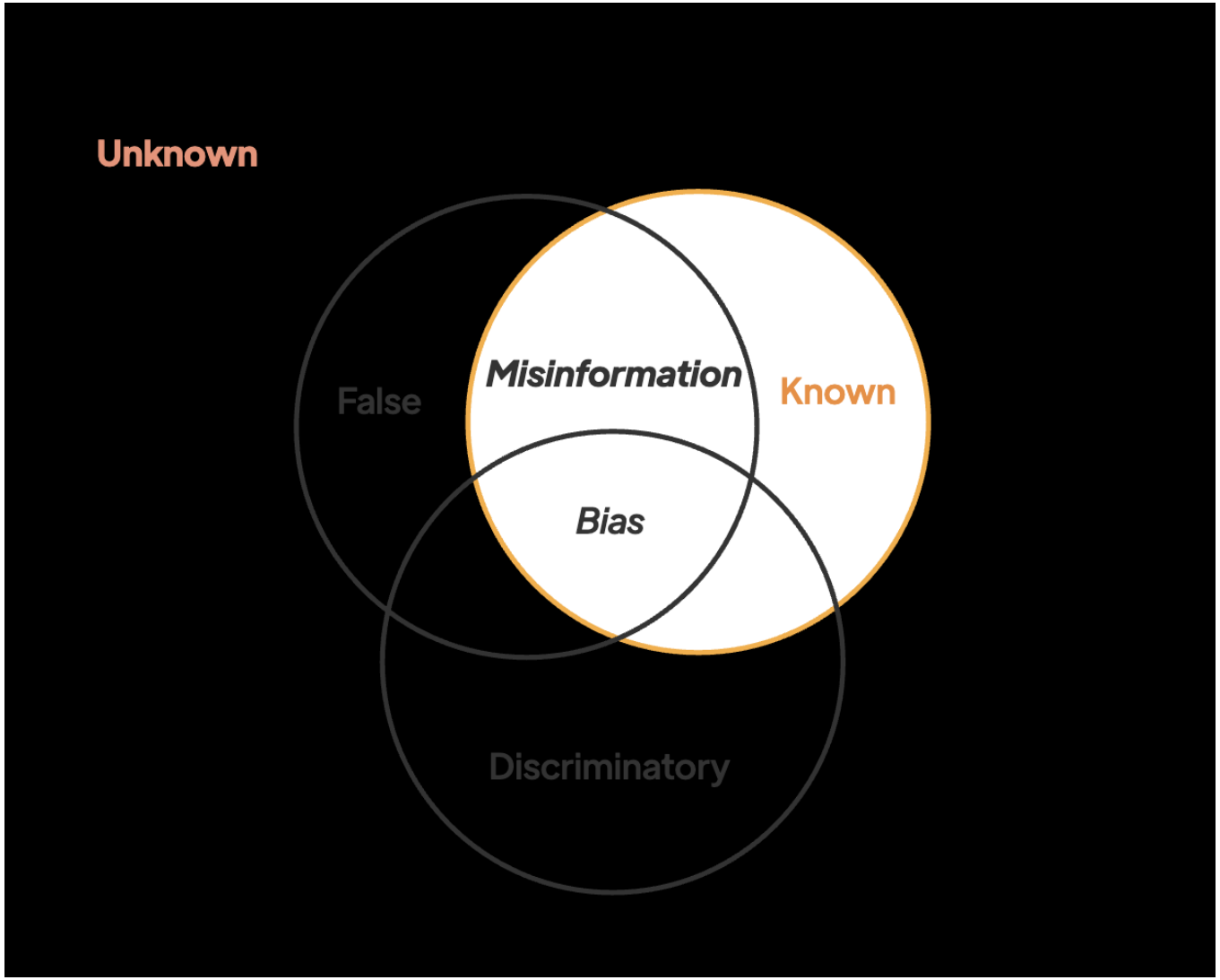


The Problem of Incomplete Information

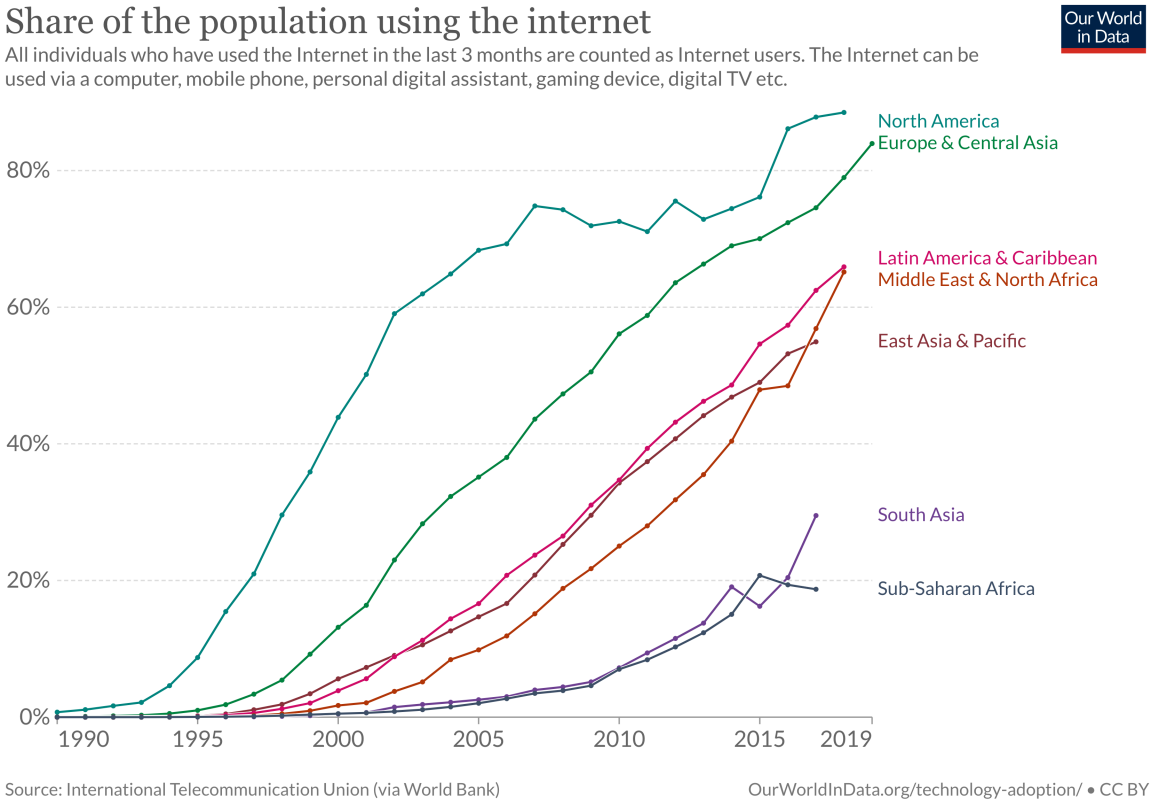
Query the Internet

↓

Incomplete Information

↓

Information ranked ~~by~~ for us



The Problem of Incomplete Information

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Information ranked ~~by~~ for us

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Incomplete Information

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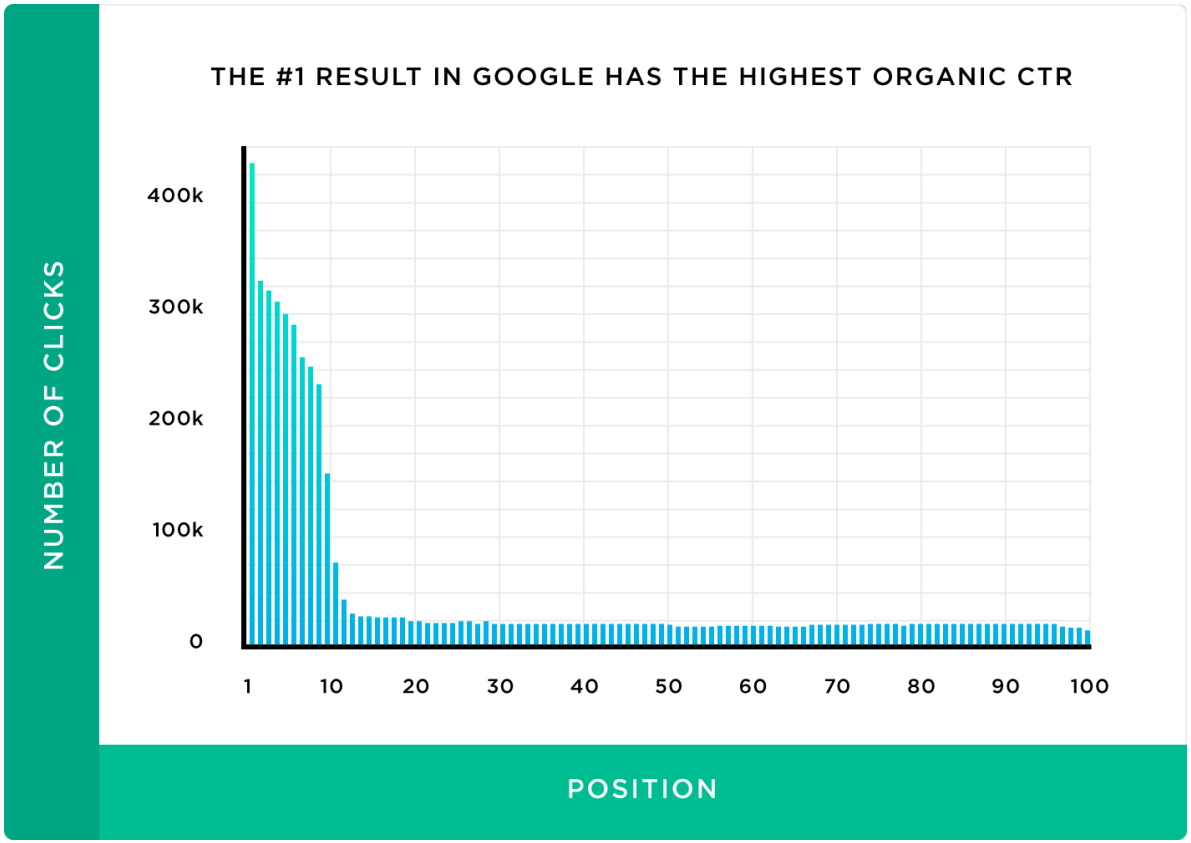
We consume the tip of the iceberg



Taken together

- The Internet is our primary source of knowledge
- We have been mapping what we see to an elusive ground truth, but have not assessed what we do *not* see
- Uncritically and consistently consuming the tip of a pre-ranked iceberg
- Harms of representation

If you control the flow of information in a society, you can influence its shared sense of right and wrong.

– Susskind, Future Politics (2018)

Objective

Knowing unknowns in an age of incomplete information[4]

Specifically, when accessing ranked information on the Internet:

1. Define metrics for information completeness
2. Understand implications of awareness of information completeness on human behavior

[4] Old question (Plato 399 BC, Einstein 1931, Taleb 2007) but we have the data and methods now to *approximate* an answer.

Metrics for Information Completeness

Defining Metrics

Balancing Relevance and Completeness

Relevance of a document given a query can be computed as the semantic distance between them in the embedding space (Microsoft DSSM, 2020).

$$\text{Query } (q) \xrightarrow[\checkmark]{\text{What I want}} \text{Search result } (r_i)$$

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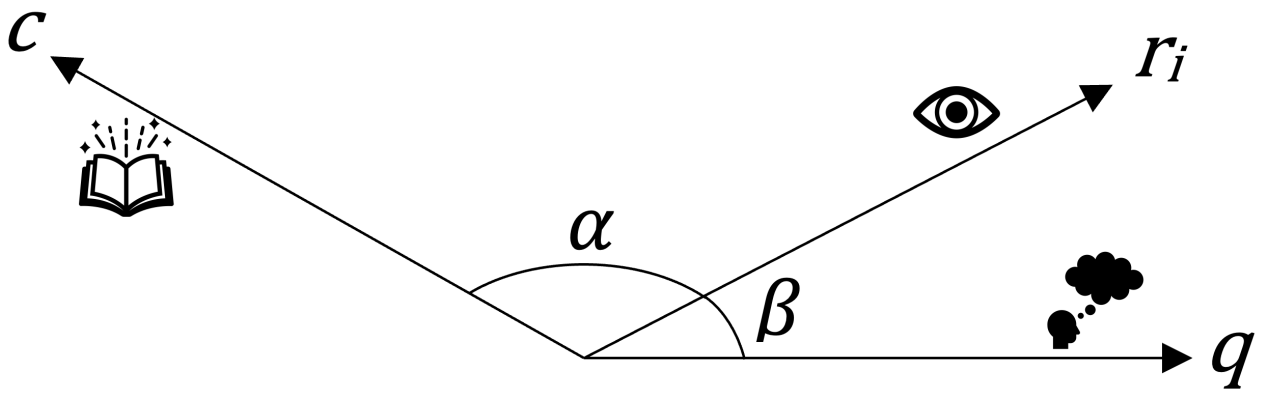
What about *completeness*?

$$\text{Query } (q) \xrightarrow[\checkmark]{\text{What I want}} \text{Search result } (r_i) \xleftarrow[\times]{\text{What exists}} \mathit{Corpus}(C)$$

Defining Completeness

Using Text Embeddings



q: query

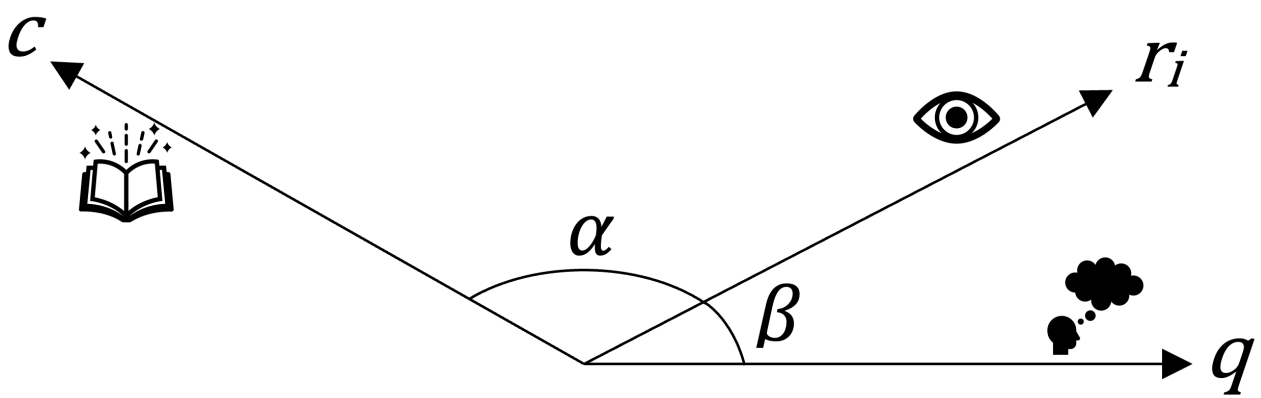
$C = \sum w_i r_i$: corpus constructed as weighted aggregate of r_i vectors

r_i: one of out n search results

w_i: weight assigned to each search result

$$I_{completeness} = \cos\alpha = \frac{\mathbf{C}\mathbf{r_i}}{\|\mathbf{C}\|\|\mathbf{r_i}\|}$$

Benefit I: Balancing Relevance and Completeness



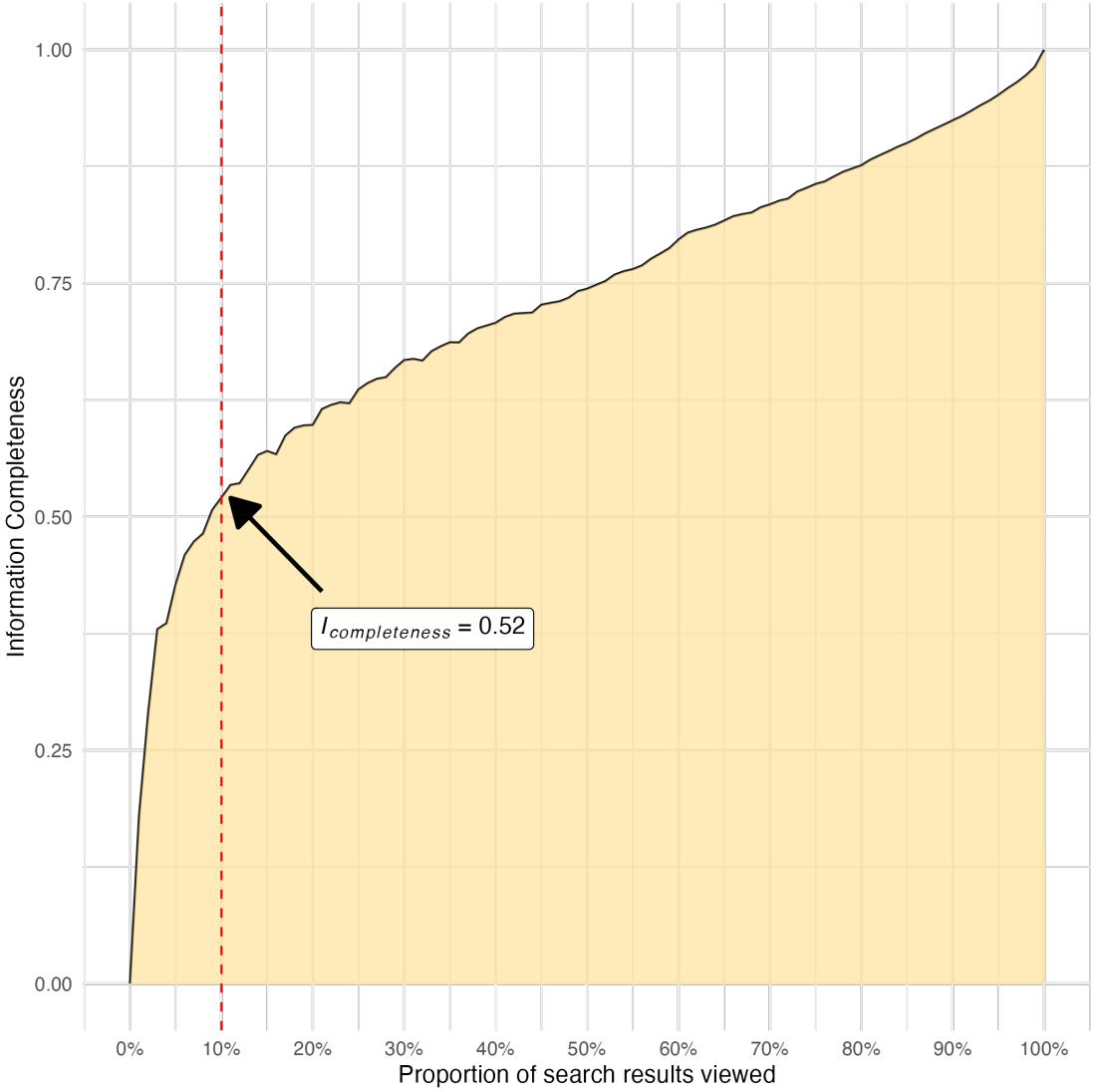
On a similar note, $I_{relevance} = \cos\beta = \frac{q\mathbf{r_i}}{\|\mathbf{q}\|\|\mathbf{r_i}\|}$

Useful as we can reorder results by their S_i score, where λ controls the balance between relevance and completeness

$$S_i = \lambda \frac{\mathbf{C}\mathbf{r_i}}{\|\mathbf{C}\|\|\mathbf{r_i}\|} + (1-\lambda)\frac{\mathbf{q}\mathbf{r_i}}{\|\mathbf{q}\|\|\mathbf{r_i}\|} = \lambda I_{completeness} + (1-\lambda) I_{relevance}$$

Benefit II: A Cumulative score per Query

Information Completeness Curves

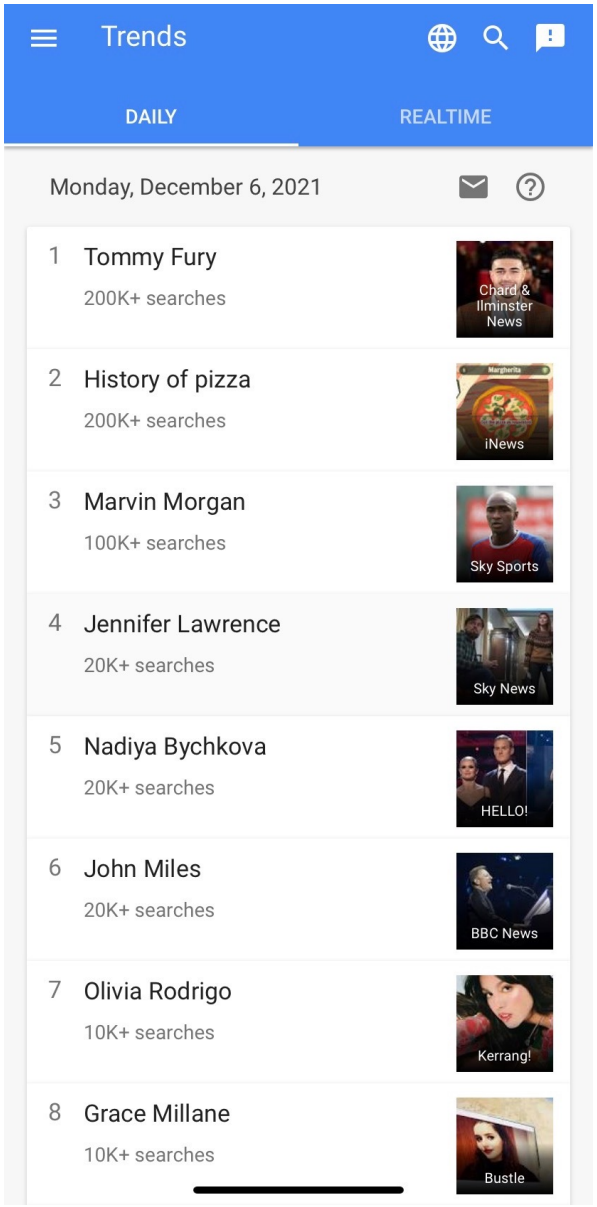


Metrics for Information Completeness

Validating Metrics

Data

Generate completeness scores for worldwide search trends.



Data

Generate completeness scores for worldwide search trends.

Everyday[5]:

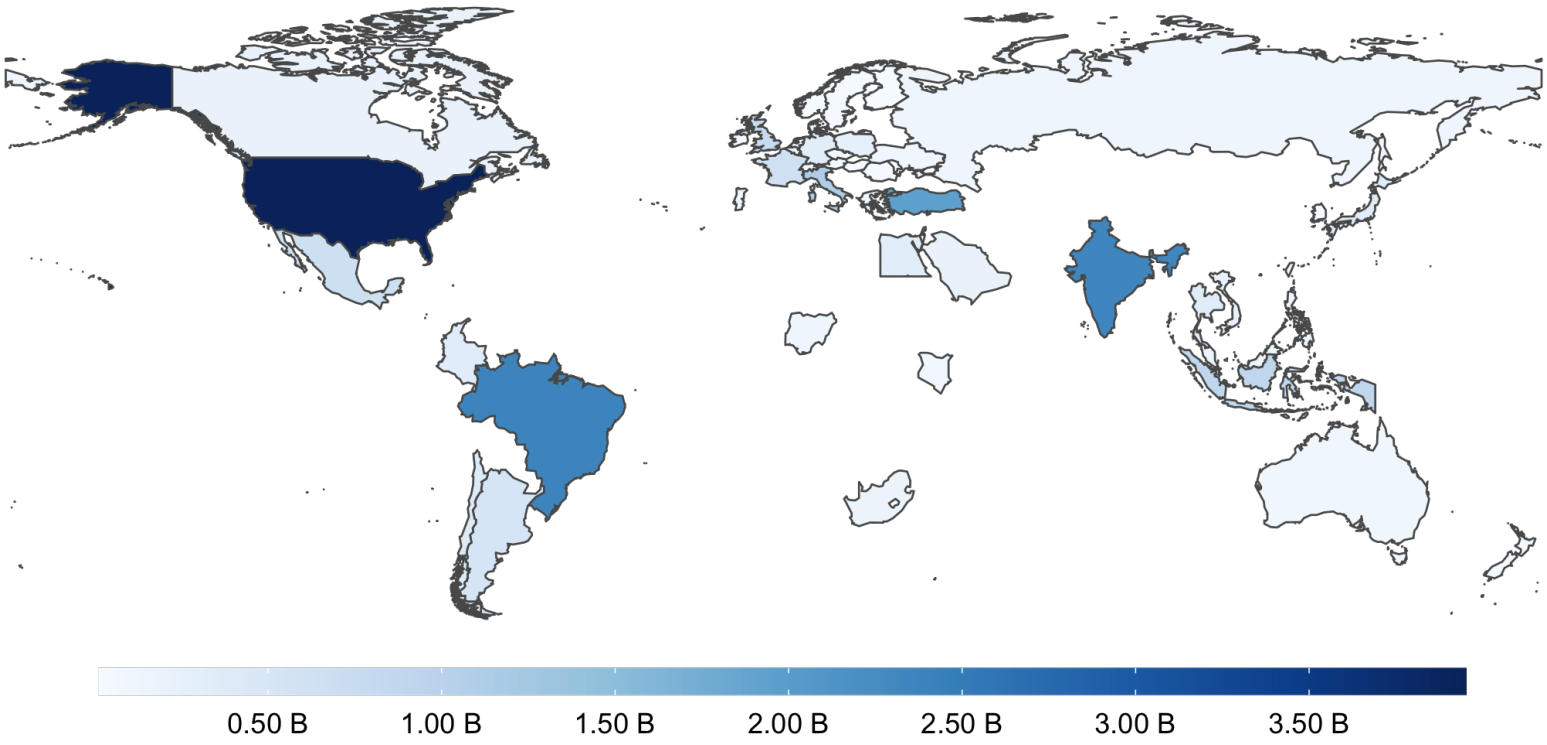
48 nations
$\times$
1.2 million searches/nation
$\times$
319 results/search[6]
$\approx$
18 billion daily data points

[5] reporting medians

[6] Both web and news search results. Since 2016, Google caps the maximum search results shown to 400.

Data

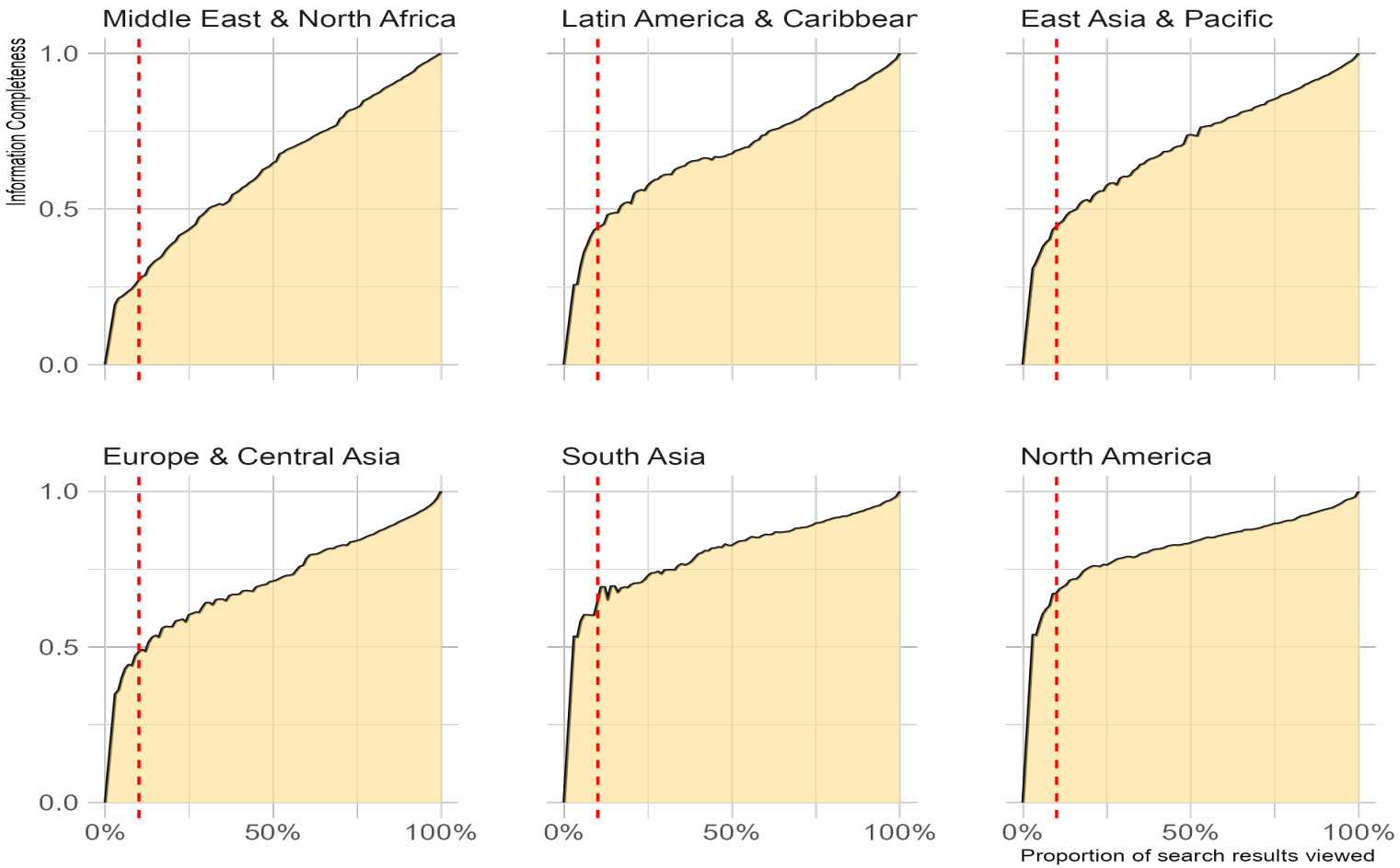
Daily Search Volume Fetched



Preliminary Results for Globally Trending Search Volume

Information Completeness Curves

Split by region, Sorted by area under the curve[7]



[7] Dashed red line marks top 10 results

Behavior Effects

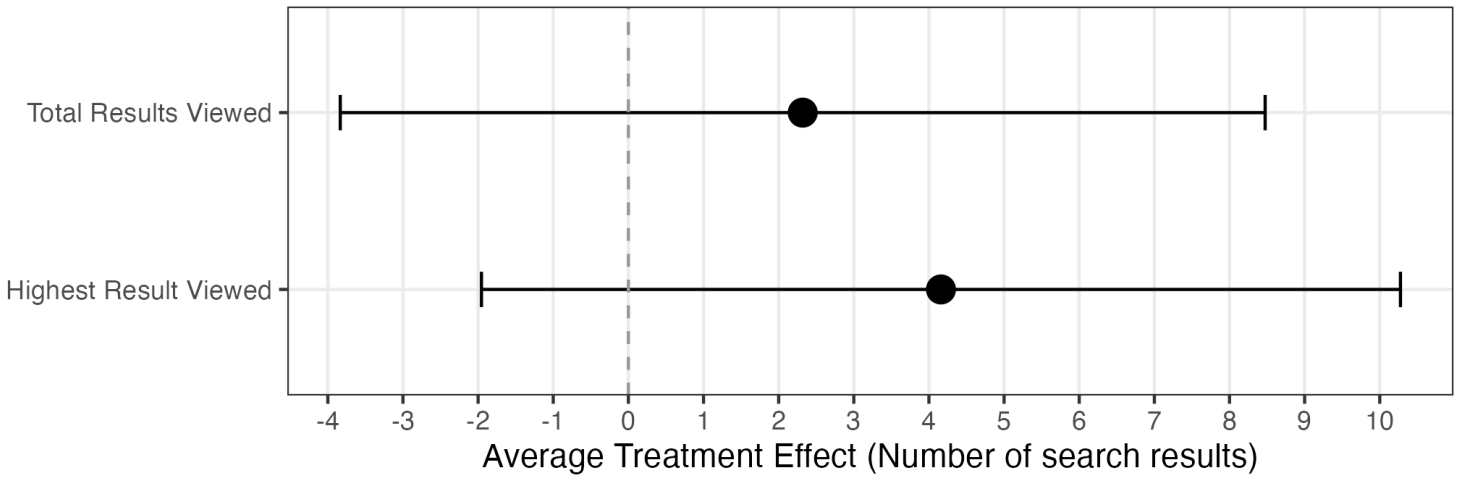
- Treatment: Viewing Information Completeness scores with search results
- Control: Viewing just search results
- Outcome:
  - How much further/more do you click?
  - AOT (Actively Open-Minded Thinking) Beliefs Scale

AOT Beliefs Scale

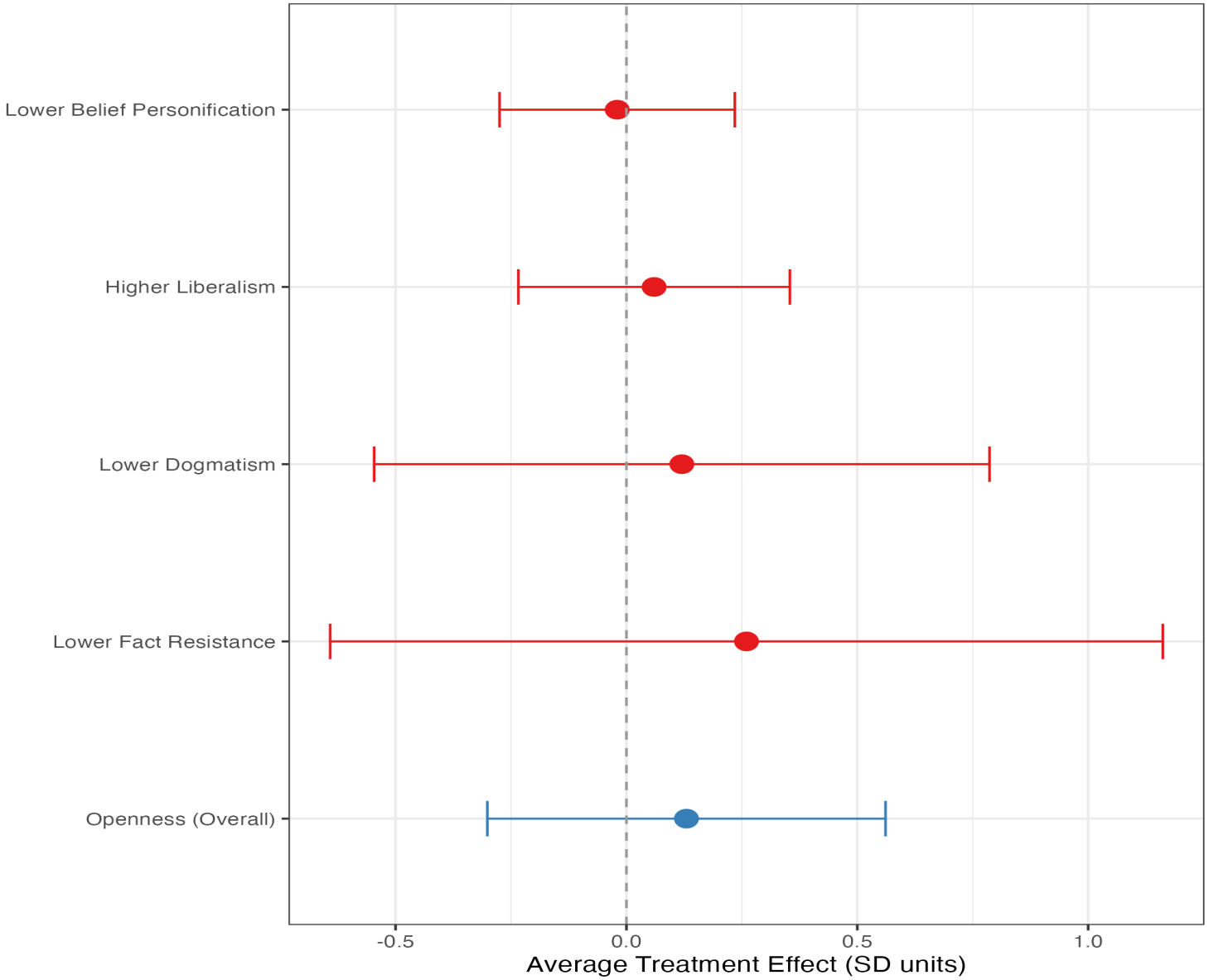
Dimension	Item
Fact resistance	One should disregard evidence that conflicts with your established beliefs. (R)
	It is important to persevere in your beliefs even when evidence is brought to bear against them. (R)
	Certain beliefs are just too important to abandon no matter how good a case can be made against them. (R)
	Beliefs should always be revised in response to new information or evidence.
	People should always take into consideration evidence that goes against their beliefs.
Dogmatism	I believe that loyalty to one's ideals and principles is more important than "open-mindedness". (R)
	I believe that the "new morality" of permissiveness is no morality at all. (R)
	Of all the different philosophies which exist in the world there is probably only one which is correct. (R)
	I think there are many wrong ways, but only one right way, to almost anything. (R)
	I believe letting youth hear controversial speakers can only confuse and mislead them. (R)
	I believe we should look to our religious authorities for decisions on moral issues. (R)
Liberalism	I consider myself broad-minded and tolerant of other people's lifestyles.
	A person should always consider new possibilities.
	I believe that the different ideas of right and wrong that people in other societies have may be valid for them.
Belief personification	There are a number of people I have come to hate because of the things they stand for. (R)
	I tend to classify people as either for me or against me. (R)
	My blood boils over whenever a person stubbornly refuses to admit he's wrong. (R)

Each survey item scored $\in \{-3, -2, -1, 1, 2, 3\}$

Effects on click behaviour



Effects on open-mindedness



Reflections

- As we consume the tip of a pre-ranked iceberg, it is crucial to assess how much we miss out on
- Sampling implications
  - Digital information is *not* sorted by the information dimension you care for. Sample ~~top *n* results~~ results until a threshold completeness is reached
- Regional differences
- Behavioral implications
- Trends Data Limitations
  - Not capturing the long tail of non-trending search queries
  - Not capturing information that was not indexed for web search

Feedback/Questions